PHYS4055 Week 2 Tutorial Questions

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Question 1: Beam-splitter transformation

Consider two independent bosonic modes with annihilation operators $\hat{a}$ and $\hat{b}$, satisfying

$$[\hat{a}, \hat{a}^\dagger] = [\hat{b}, \hat{b}^\dagger] = 1, \qquad [\hat{a}, \hat{b}] = [\hat{a}, \hat{b}^\dagger] = 0.$$

A lossless beam splitter is described by the unitary operator

$$\hat{U}_B = \exp\left[\theta\left(\hat{a}^\dagger\hat{b} - \hat{a}\hat{b}^\dagger\right)\right].$$

The Baker–Campbell–Hausdorff relation may be written as

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!}[\hat{A}, [\hat{A}, \hat{B}]] + \frac{1}{3!}[\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]] + \cdots .$$

Using this relation:

1. Calculate the first four nested commutators required to evaluate

$$\hat{U}_B\hat{a}\hat{U}_B^\dagger \quad \text{and} \quad \hat{U}_B\hat{b}\hat{U}_B^\dagger.$$

2. Group the resulting series into sine and cosine expansions and show that

$$\hat{U}_B\hat{a}\hat{U}_B^\dagger = \hat{a}\cos\theta - \hat{b}\sin\theta,$$

$$\hat{U}_B\hat{b}\hat{U}_B^\dagger = \hat{a}\sin\theta + \hat{b}\cos\theta.$$

3. Defining the beam-splitter transmissivity as

$$\eta = \cos^2\theta,$$

show that the output-mode operators are

$$\hat{a}' = \sqrt{\eta}\,\hat{a} - \sqrt{1-\eta}\,\hat{b},$$

$$\hat{b}' = \sqrt{1-\eta}\,\hat{a} + \sqrt{\eta}\,\hat{b}.$$

A. Question 2: Energy conservation at a beam splitter

Consider a lossless beam splitter acting on two optical modes. The input and output mode operators are related by

$$\hat{a}' = \sqrt{\eta}\,\hat{a} - \sqrt{1-\eta}\,\hat{b},$$

and

$$\hat{b}' = \sqrt{1-\eta}\,\hat{a} + \sqrt{\eta}\,\hat{b},$$

where $0 \le \eta \le 1$ is the beam-splitter transmissivity.

The photon-number operators of the two output modes are

$$\hat{N}_{a'} = \hat{a}'^\dagger\hat{a}', \qquad \hat{N}_{b'} = \hat{b}'^\dagger\hat{b}'.$$

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1. Write down the transformed creation operators $\hat{a}'^{\dagger}$ and $\hat{b}'^{\dagger}$.
2. Calculate $\hat{N}_{a'}$ explicitly in terms of the input-mode operators $\hat{a}$, $\hat{a}^{\dagger}$, $\hat{b}$, and $\hat{b}^{\dagger}$.
3. Similarly, calculate $\hat{N}_{b'}$ in terms of the input-mode operators.
4. By adding your results, show that

$$\hat{N}_{a'} + \hat{N}_{b'} = \hat{N}_a + \hat{N}_b,$$

where

$$\hat{N}_a = \hat{a}^{\dagger}\hat{a}, \qquad \hat{N}_b = \hat{b}^{\dagger}\hat{b}.$$

B. Question 3: Partial trace

A single photon passes through a balanced beam splitter, producing the two-mode state

$$|\psi\rangle_{ab} = \frac{1}{\sqrt{2}} \left(|1\rangle_a |0\rangle_b + |0\rangle_a |1\rangle_b \right).$$

1. Write down the two-mode density operator

$$\hat{\rho}_{ab} = |\psi\rangle_{ab}\langle\psi|.$$

2. Suppose that we only have access to mode a. Calculate its reduced density operator by tracing over mode b:

$$\hat{\rho}_a = \mathrm{Tr}_b(\hat{\rho}_{ab}) = \sum_{n=0}^{1} {}_b\langle n|\hat{\rho}_{ab}|n\rangle_b.$$

3. What is the probability of detecting zero photons in mode a? What is the probability of detecting one photon?
4. Is the reduced state $\hat{\rho}_a$ pure or mixed?